TREES AND THEIR DEGREES

MARIE-CAMILLE DELARUE

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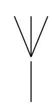
1. Introduction

In the course of our study of the family of Higman–Thompson groups, there came to be a question that was very easy to answer in small dimensions and much harder in higher dimensions. The fundamental reason for this was that it was easy to come up with a reasonable visualization in low dimensions, and much harder in higher dimensions because of the limitations of 3-dimensional space. Therefore, we tried to come up with a good visualization in higher dimensions.

The following document explains this problem and the framework we came up with to answer it in higher dimensions. For extra context on the Thompson groups and their homology, please consult the other file in this folder. Reading the following document is however completely necessary to understanding the specific question we were working on as well as its implementation.

2. Trees

We will consider finite rooted trees, and we call an a**-junction** a vertex of the tree with one parent and a children: For instance,  is a 3-junction.

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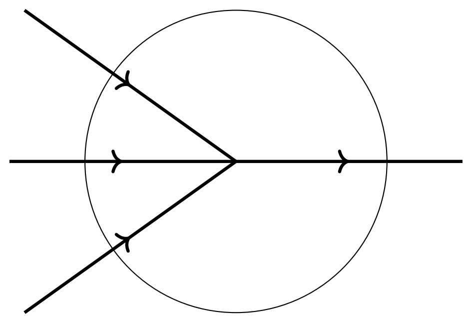
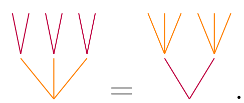
2.1. **Trees with simple junctions.** Let us start by considering the following problem. Consider a finite rooted tree which only has a-junctions, for a an integer. We will think of trees as being oriented from the leaves to the root. Consider the following picture:



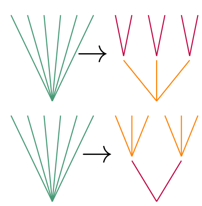
We will call the **degree** of a junction the number of incoming edges minus the number of outgoing edges in a small circle around the junction. For any a-junction, this number is clearly $a-1$.

Now let us assume we have trees with a-junctions and b-junctions. The a-junctions have degree $a-1$, and the b-junctions have degree $b-1$.

2.2. **Trees with double junctions.** Now let us consider a family of trees with both a-junctions and b-junctions, for a and b two different integers, with the following additional **interchange relation**:



Each a-junction has degree $a-1$ and each b-junction has degree $b-1$ again, but what happens when the junctions are switching? We add an ab-junction which represents the switch:



Now we've added an extra junction, and we would like to determine its degree. It is tempting to say that the degree of this junction is $ab-1$. However, this does not really work, in part because our definition of degree is not precise enough.

Let's assume we have an ab-junction of this form:

In a neighborhood of this ab-junction however, we have two very different pictures. So if we move just a little bit away from the ab-junction, we can end up in two very different scenarios which have different degrees.

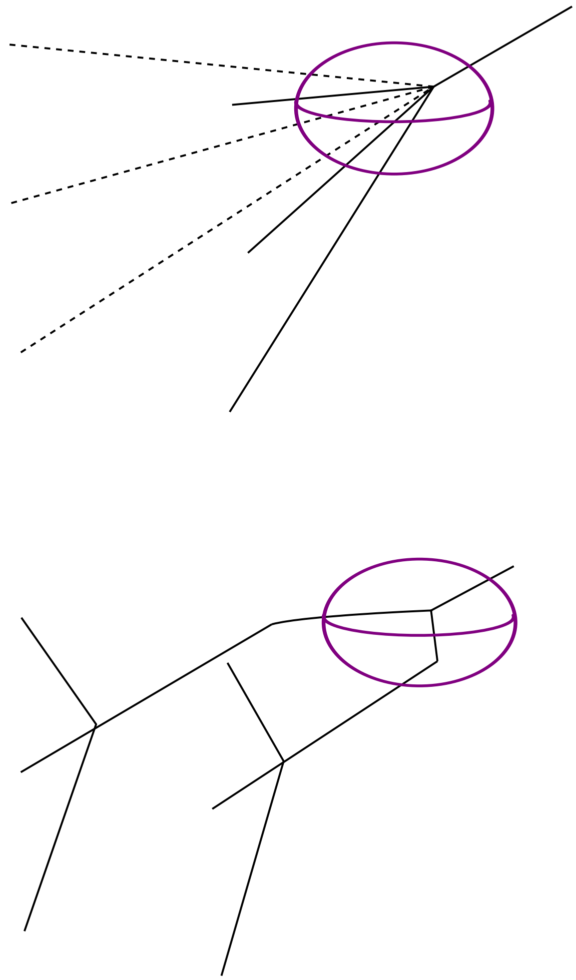


FIGURE 1. Two trees very close to each other but with very different "local images". The trees are drawn in the direction they are oriented.

2.3. **New degree definition.** Let's give another definition of the degree by going back to the case of a simple junction. If we imagine a smaller lens traveling around the junction and recording what it sees, we will see a single path $a+1$ times: a time with positive orientation and once with negative orientation, and so we recover a degree of $a-1$.

We will find the degree of the ab-junction by traveling around it and recording what we see. Because the ab-junction lives in "higher dimension", we need to count the number of a-junctions and b-junctions that we see around it, not just paths.

Explain this better. Also I should try to make a picture for the bigger junctions but I'm just not sure how

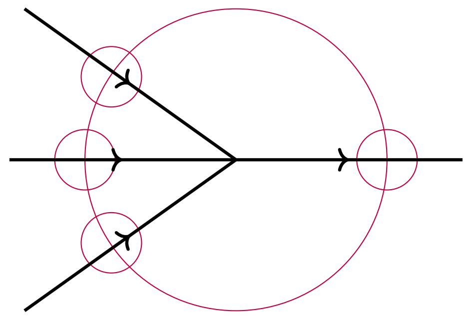


FIGURE 2. The case of a simple junction

This is the basis of the heuristic we will follow.